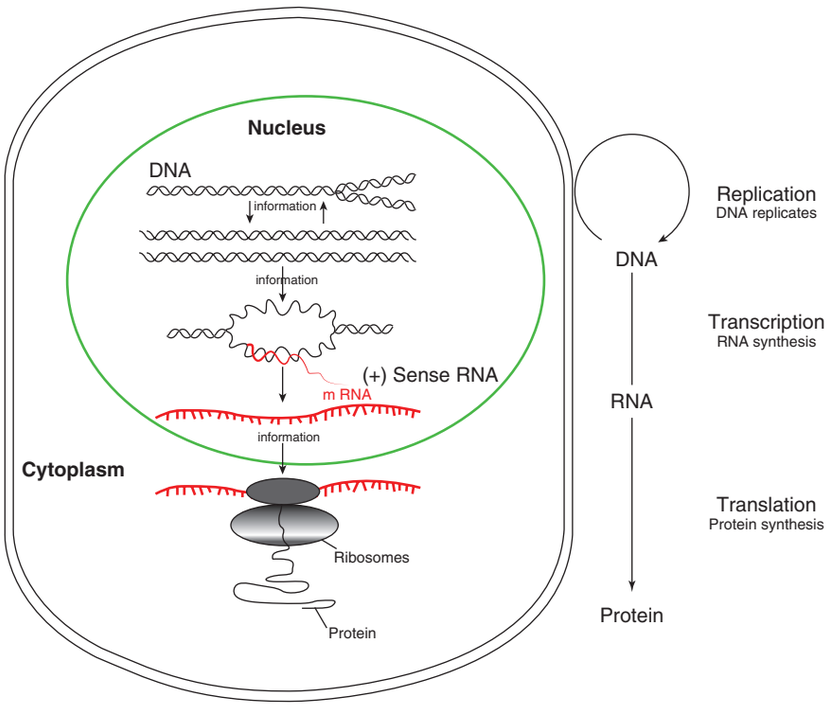


Figure 12.1: Central dogma of molecular biology

copying of parental DNA to form daughter DNA molecules with identical nucleotide sequences. The information contained in the base sequence of DNA is copied into protein molecule through an RNA molecule. The second is **transcription**, production of mRNA from DNA. It is the process by which the segment corresponding to a particular gene is selected and an RNA molecule is synthesized. The third is **translation**, The production of an amino acid sequence from an RNA base sequence. The genetic message encoded in messenger RNA (mRNA) is translated on the ribosomes into a polypeptide with a particular sequence of amino acids. The order of amino acid in a polypeptide chain is determined by DNA base sequence.

12.2 Transcription

An important feature of RNA synthesis is that even though the DNA molecule being copied is double stranded, in any particular region of DNA only one strand serves as a template. The DNA strand copied into RNA molecule is called CODING OR SENSE STRAND.

The synthesis of RNA consists of five discrete stage (Figure 12.2):

1. Promoter recognition: RNA polymerase binds to DNA within a specific base sequence (20–200 bases long) called a promoter. The sequence TATAAT (or a nearly identical sequence) often called a pribnow box or – 10 region is found as part of all prokaryotic promoters.

   The RNA polymerase of the bacterium *E.coli* consists of five protein subunits. Four of the subunits comprise the core enzyme (catalyzes the joining of the nucleoside triphosphates to the RNA) and fifth subunit, the σ subunit (required for promoter binding).
2. Local unwinding of DNA occurs and RNA polymerase forms an open promoter complex.

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